

Network Administration / System Administration

Homework 5

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1 Setting up PowerDNS

1.1 Install Debian and PowerDNS

Reference:

- <https://wiki.debian.org/QEMU>
- [How to add user to sudoers in Debian](#)
- [Bash: usermod command not found in latest Debian](#)
- [Install PowerDNS and PowerDNS Admin on Debian](#)
- <https://repo.powerdns.com/>
- <https://doc.powerdns.com/authoritative/installation.html>
- [PowerDNS 架設 - Debian](#)

1. 我連線到工作站上，並且下載 Debian 12.10.0 的 [netinst.iso](#) 安裝檔案。

2. 使用 qemu-img 建立一個 16GB 的 qcow2 磁碟檔案。

```
1 qemu-img create -f qcow2 debian.qcow 16G
```

3. 使用 qemu-system-x86_64 開啟虛擬機器。連線到 port 7000 後，安裝 Debian 12。

```
1 qemu-system-x86_64 \  
2   -enable-kvm \  
3   -m 2048 \  
4   -smp 2 \  
5   -cpu host \  
6   -cdrom debian-12.10.0-amd64-netinst.iso \  
7   -drive file=debian.qcow2 \  
8
```

```
8 -boot d \  
9 -net nic -net user \  
10 -vnc :1100
```

4. 在安裝精靈當中，我設定 Hostname 為 debian，User name、Root password 以及 User password 為我的學號，Debian Archive Mirror 為 `debian.csie.ntu.edu.tw`，並且選擇安裝 gnome desktop、SSH server 和 Standard system utilities。其餘選項以及 Grub 設定我都使用預設值。

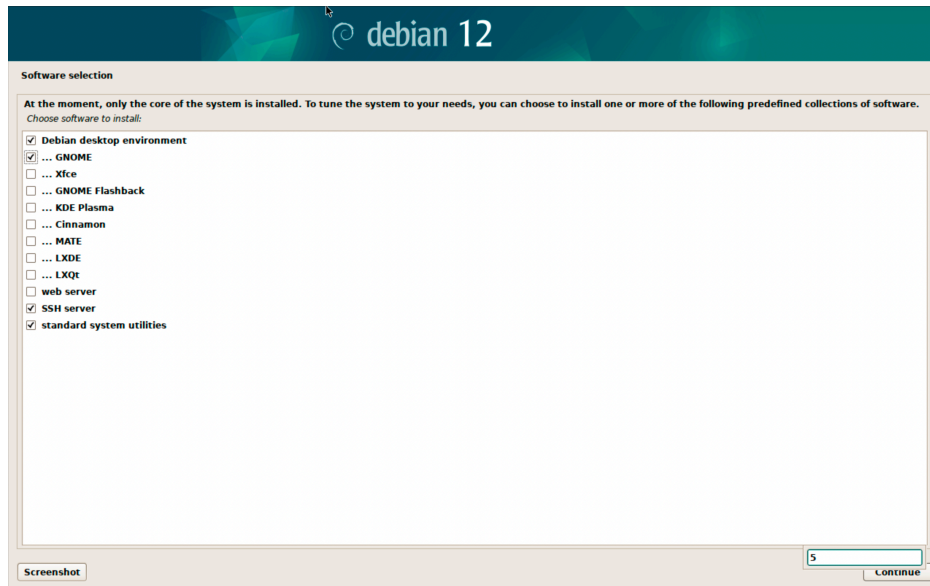


Figure 1: Debian 安裝精靈

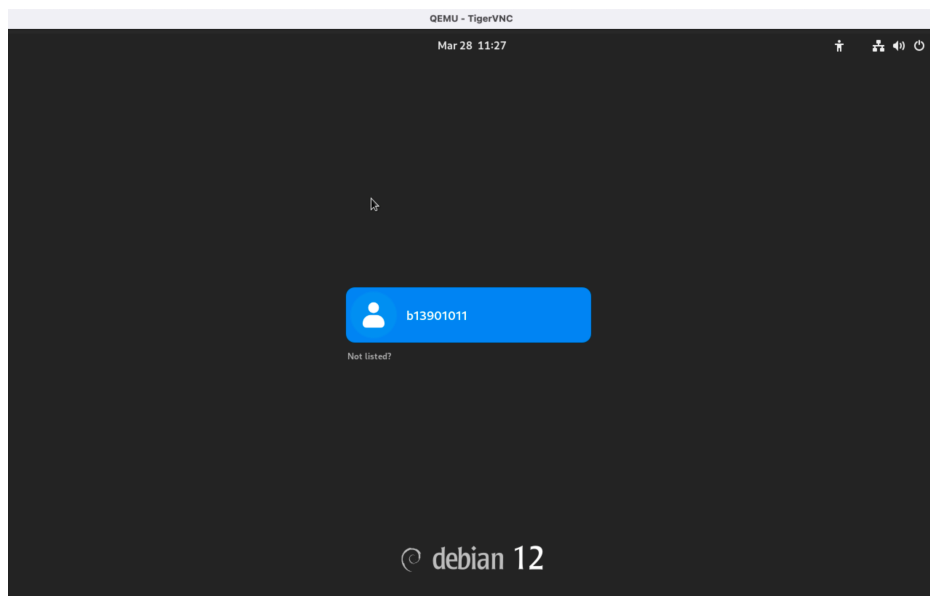


Figure 2: 完成 Debian 安裝

5. 安裝完成後，我關閉虛擬機器，並且參考 Homework 2，建立一份 `run_vm.sh` 作為開啟虛擬機器的 Shell Script。這個腳本會開啟虛擬機器，並且將 VNC 連線的 port 設定為 7000，SSH 的 port 設定為 4321。在 `change vnc password` 之後，就可以透過 VNC 連線到虛擬機器，並且使用 SSH 連線到虛擬機器。

```
1 #!/bin/bash
2 qemu-system-x86_64 \
3     -enable-kvm \
4     -vnc :1100, password=on \
5     -monitor stdio \
6     -m 2048 \
7     -smp 2 \
8     -cpu host \
9     -drive file=debian.qcow2 \
10    -boot d \
11    -net nic -net user,hostfwd=tcp::4321-:22
```

6. 在工作站上，使用 `ssh b13901011@localhost -p 4321` 連線到虛擬機器，並且輸入 `su -` 進入 root 權限。

7. 執行以下指令安裝必要的套件。

```
1 apt update
2 apt upgrade -y
3 apt install -y sudo vim git curl wget net-tools libpq-dev neofetch
```

8. 將我的使用者加入 `sudo` 群組。

```
1 usermod -aG sudo b13901011
```

9. 安裝 MariaDB。

```
1 sudo apt install mariadb-server mariadb-client
2 sudo systemctl start mariadb
3 sudo systemctl enable mariadb
```

10. 以 `sudo mysql -u root` 登入 MariaDB，並且執行以下指令：

```
1 CREATE DATABASE powerdns;
2 GRANT ALL ON powerdns.* TO 'powerdns_user'@'%' IDENTIFIED BY 'b13901011';
3 FLUSH PRIVILEGES;
4 EXIT;
```

11. 停止 systemd-resolved 服務

```
1 sudo systemctl stop systemd-resolved
2 sudo systemctl disable systemd-resolved
```

12. 修改 DNS 設定檔案 /etc/resolv.conf 的內容：

```
1 sudo unlink /etc/resolv.conf
2 sudo vim /etc/resolv.conf
```

```
1 # /etc/resolv.conf
2 nameserver 8.8.8.8
3 nameserver 8.8.4.4
```

13. 加上指定版本的 Repository：

修改 /etc/apt/sources.list.d/pdns.list 的內容，新增如下：

```
1 deb [signed-by=/etc/apt/keyrings/auth-49-pub.asc]
   ↪ http://repo.powerdns.com/debian bookworm-auth-49 main
```

修改 /etc/apt/preferences.d/auth-49 的內容，新增如下：

```
1 Package: auth*
2 Pin: origin repo.powerdns.com
3 Pin-Priority: 600
```

並且安裝 PowerDNS。

```
1 sudo install -d /etc/apt/keyrings;
2 curl https://repo.powerdns.com/FD380FBB-pub.asc | sudo tee
   ↪ /etc/apt/keyrings/auth-49-pub.asc &&
3 sudo apt-get update &&
4 sudo apt-get install pdns-server pdns-backend-mysql
```

14. Import PowerDNS database schemas to MariaDB

```
1 mysql -u powerdns_user -p powerdns <
   ↪ /usr/share/pdns-backend-mysql/schema/schema.mysql.sql
```

```

MariaDB [powerdns]> show tables;
+-----+
| Tables_in_powerdns |
+-----+
| comments            |
| cryptokeys          |
| domainmetadata      |
| domains             |
| records             |
| supermasters        |
| tsigkeys            |
+-----+
7 rows in set (0.000 sec)

```

Figure 3: powerdns database schemas

15. 修改 PowerDNS 設定檔案 `/etc/powerdns/pdns.conf` 的內容，新增如下：

```

1 #####
2 # local-port      The port on which we listen
3 #
4 local-port=5301

```

16. 新增一個檔案 `/etc/powerdns/pdns.d/pdns.local.gmysql.conf` 存放 powerdns 的 database 相關設定，內容如下：

```

1 # MySQL Configuration
2 # Launch gmysql backend
3 launch+=gmysql
4 # gmysql parameters
5 gmysql-host=127.0.0.1
6 gmysql-port=3306
7 gmysql-dbname=powerdns
8 gmysql-user=powerdns_user
9 gmysql-password=b13901011
10 gmysql-dnssec=yes
11 # gmysql-socket=

```

17. 修改設定檔案的權限：

```

1 sudo chown pdns: /etc/powerdns/pdns.d/pdns.local.gmysql.conf
2 sudo chmod 640 /etc/powerdns/pdns.d/pdns.local.gmysql.conf

```

18. 測試 PowerDNS 是否能正常運作：

```
1 sudo systemctl stop pdns.service
2 sudo pdns_server --daemon=no --guardian=no --loglevel=9
```

```
root@debian:/etc/powerdns# sudo systemctl stop pdns.service
sudo pdns_server --daemon=no --guardian=no --loglevel=9
Mar 29 12:25:55 Loading '/usr/lib/x86_64-linux-gnu/pdns/libbindbackend.so'
Mar 29 12:25:55 [bindbackend] This is the bind backend version 4.9.4 (Feb  6 2025 15:18:23) (with bind-dnssec-db support) reporting
Mar 29 12:25:55 Loading '/usr/lib/x86_64-linux-gnu/pdns/libmysqlbackend.so'
Mar 29 12:25:55 [mysqlbackend] This is the mysql backend version 4.9.4 (Feb  6 2025 15:18:23) reporting
Mar 29 12:25:55 This is a standalone pdns
Mar 29 12:25:55 Created local state directory '/var/run/pdns/'
Mar 29 12:25:55 Listening on controlsocket in '/var/run/pdns/pdns.controlsocket'
Mar 29 12:25:55 UDP server bound to 0.0.0.0:5301
Mar 29 12:25:55 UDP server bound to [::]:5301
Mar 29 12:25:55 TCP server bound to 0.0.0.0:5301
Mar 29 12:25:55 TCP server bound to [::]:5301
Mar 29 12:25:55 PowerDNS Authoritative Server 4.9.4 (C) PowerDNS.COM BV
Mar 29 12:25:55 Using 64-bits mode. Built using gcc 12.2.0 on Feb  6 2025 15:18:23 by root@localhost.
Mar 29 12:25:55 PowerDNS comes with ABSOLUTELY NO WARRANTY. This is free software, and you are welcome to redistribute it according to the terms of the GPL version 2.
Mar 29 12:25:55 [stub-resolver] Doing stub resolving for 'auth-4.9.4.security-status.secpoll.powerdns.com.[TXT]', using resolvers: 8.8.8.8, 8.8.4.4
Mar 29 12:25:56 [stub-resolver] Question for 'auth-4.9.4.security-status.secpoll.powerdns.com.[TXT]' got answered by 8.8.8.8
Mar 29 12:25:56 Polled security status of version 4.9.4 at startup, no known issues reported: OK
Mar 29 12:25:56 [bindbackend] Parsing 0 domain(s), will report when done
Mar 29 12:25:56 [bindbackend] Done parsing domains, 0 rejected, 0 new, 0 removed
Mar 29 12:25:56 mysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 Creating backend connection for TCP
Mar 29 12:25:56 mysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 About to create 3 backend threads for UDP
Mar 29 12:25:56 mysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 mysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 mysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 Done launching threads, ready to distribute questions
```

Figure 4: PowerDNS 測試

19. 確認沒問題後，重新啟動 PowerDNS 服務，並且截圖題目要求的畫面。

```
1 sudo systemctl restart pdns
2 sudo systemctl enable pdns
```

Answer:

```
root@debian:/etc/powerdns# dig @localhost -p 5301

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> @localhost -p 5301
; (2 servers found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: REFUSED, id: 8687
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;.                               IN      NS

;; Query time: 0 msec
;; SERVER: ::1#5301(localhost) (UDP)
;; WHEN: Sat Mar 29 12:31:46 CST 2025
;; MSG SIZE rcvd: 28

root@debian:/etc/powerdns# pdns_control version
4.9.4
```

Figure 5: dig @localhost -p 5301 以及 pdns_control version 結果

1.2 Install PowerDNS Admin

Reference:

- [Install PowerDNS and PowerDNS Admin on Debian](#)
- <https://docs.docker.com/engine/install/debian/#install-using-the-repository>
- <https://github.com/PowerDNS-Admin/PowerDNS-Admin>
- <https://github.com/PowerDNS-Admin/PowerDNS-Admin/issues/816#issuecomment-809533428>

1. 我打算根據官網 README 的指示，使用 Docker 安裝 PowerDNS Admin。首先，我在虛擬機器上安裝 Docker。

2. 根據 Docker 的官方說明，我使用以下指令安裝 Docker：

```

1 # Add Docker's official GPG key:
2 sudo apt-get update
3 sudo apt-get install ca-certificates curl
4 sudo install -m 0755 -d /etc/apt/keyrings
5 sudo curl -fsSL https://download.docker.com/linux/debian/gpg -o
   ↪ /etc/apt/keyrings/docker.asc
6 sudo chmod a+r /etc/apt/keyrings/docker.asc
7 # Add the repository to Apt sources:
8 echo \
9     "deb [arch=$(dpkg --print-architecture)
   ↪ signed-by=/etc/apt/keyrings/docker.asc]
   ↪ https://download.docker.com/linux/debian \
10     $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
11     sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
12
13 sudo apt-get update
14 sudo apt-get install docker-ce docker-ce-cli containerd.io
   ↪ docker-buildx-plugin docker-compose-plugin

```

3. 將我的使用者加入 Docker 群組。

```

1 sudo usermod -aG docker $USER
2 newgrp docker

```

4. Clone PowerDNS Admin 的 GitHub repository。

```

1 git clone https://github.com/PowerDNS-Admin/PowerDNS-Admin.git
2 cd PowerDNS-Admin

```

5. 使用 `ip addr` 獲取 Debian VM 的 IP 位址為 10.0.2.15

6. 修改 PowerDNS 的設定檔：編輯 `/etc/powerdns/pdns.conf` 的內容，新增如下：

```
1 api=yes
2 api-key=b13901011
3 webserver=yes
4 webserver-address=10.0.2.15 # your real ip not localhost
5 webserver-allow-from=127.0.0.1,192.168.0.1,172.17.0.2
```

7. 重新啟動 PowerDNS 服務：

```
1 sudo systemctl restart pdns
```

8. 使用以下指令啟動 PowerDNS Admin：

```
1 docker run -d \
2   -e SECRET_KEY='a-very-secret-key' \
3   -v pda-data:/data \
4   -p 9191:80 \
5   powerdnsadmin/pda-legacy:latest
```

9. 我使用 vnc 連線到虛擬機器，並且使用瀏覽器開啟 `http://localhost:9191`，然後註冊一組帳號密碼，並且登入 PowerDNS Admin。
10. 登入 PowerDNS Admin 之後，會進入 Server Settings 頁面，在左側 Settings Editor 當中，PowerDNS Server 填寫 `http://10.0.2.15:8081/`，API Key 填寫 `b13901011`，PowerDNS Version 填寫 `4.9.4`，然後按下 Save 按鈕。之後，我進到 Profile 頁面，並截圖。

Answer:

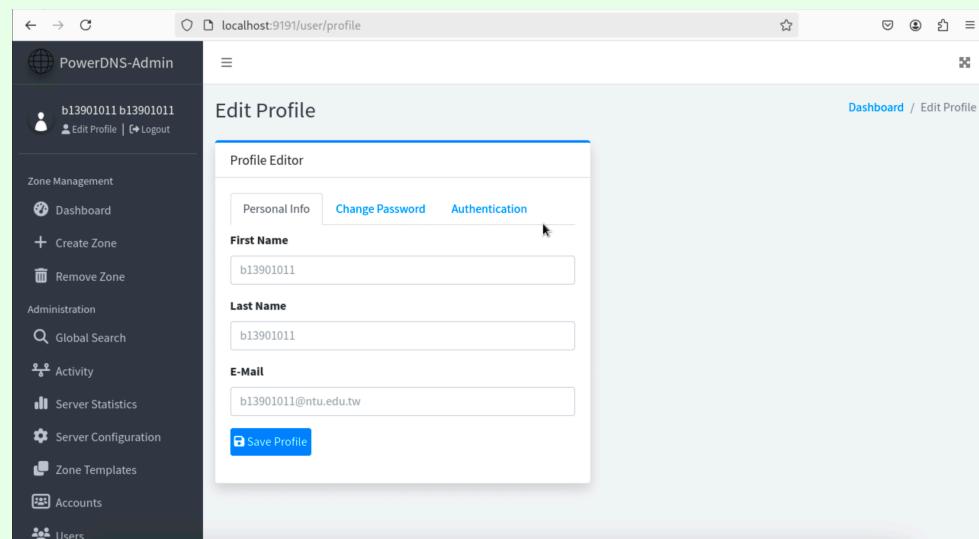


Figure 6: PowerDNS Admin Profile